

AN INTRODUCTION TO ARTIN L -FUNCTIONS

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An Artin L -function is a generalization of the Riemann zeta function and the classical Dirichlet L -functions. Just as the Dirichlet L -functions are useful in the study of primes in arithmetic progressions, so are the Artin L -functions useful in the study of the decomposition of primes in algebraic number fields. In contrast to the classical objects, we still do not have analytic continuation of these objects in the general setting. If we did, this would have profound consequences in the study of prime number theory, especially to various forms of the effective Chebotarev density theorem, which can be viewed as the most general form of the prime number theorem.

It was in the 1950's that Brauer, using his celebrated induction theorem, first obtained a meromorphic continuation of the Artin L -functions. Later, in the 1960's, when Langlands outlined his general program, he formulated a stronger conjecture that Artin L -functions "arise from" automorphic representations, which is a striking instance of the theme of "modularity conjectures" that have now become central in number theory. Indeed, it can be said without exaggeration that Langlands [17] was the first to prove a non-abelian "modularity conjecture" that launched a programme to attack such questions. The most notable application of this result was to the theory of elliptic curves and thus, as a consequence to the celebrated Fermat's last theorem, which arose from earlier work of Hellegouarch, Frey, Serre, Ribet and finally, Wiles. It is equally remarkable that ideas from the latter theory are establishing new cases of "modularity" in the theory of Artin L -functions, as seen in the recent work [8] and [36] for instance.

This article is based on lectures given at the Tata Institute of Fundamental Research in December 1998. I am grateful to the Tata Institute for this opportunity as well as to A. Raghuram who latexed these lectures and corrected the mathematics in some places. The article surveys the many methods being invoked to study Artin L -functions, but focuses on the author's own work related to it, especially in the context of Selberg's conjectures. I would also like to thank NSERC and a Killam Research Fellowship for financial support.

1. MOTIVATING EXAMPLES

1.1. The Riemann ζ -function. The Riemann zeta function is a basic prototype of L -functions occurring in number theory. This function was introduced by Riemann in 1860 and is defined by

$$(1) \quad \zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} ; \quad \text{for } \sigma > 1.$$

We shall, following the number theoretic tradition started by Riemann, use the symbol $s = \sigma + it$ for a complex variable s , whose real part is σ and imaginary part is t .

The fundamental properties of the Riemann zeta function we are interested in are summarised in the following theorem. (See Section 5.1 of [28].)

- Theorem 1.1.** **1. Convergence :** *The series defining $\zeta(s)$ converges absolutely and uniformly on any compact subset of the region $\sigma \geq 1 + \epsilon$ for every $\epsilon > 0$.*
- 2. Euler Product :** *For $\sigma > 1$ we have an Euler product expansion for $\zeta(s)$ given by*

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}$$

the product being taken over all prime numbers p in $\mathbb{Z}$.

- 3. Analytic continuation & Functional equation :** *The function $\zeta(s)$ has a meromorphic continuation to the half plane $\sigma > 0$ and is holomorphic except for a simple pole at $s = 1$ with residue equal to 1. Thus $(s - 1)\zeta(s)$ extends to an entire function of s . Further let*

$$\Phi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

where $\Gamma(s)$ is the usual gamma function. Then $\Phi(s)$ admits a meromorphic continuation to the entire s -plane and is holomorphic everywhere except for simple poles at $s = 0, 1$ and satisfies the functional equation

$$\Phi(s) = \Phi(1 - s).$$

From the Euler product it is clear that $\zeta(s) \neq 0$ for $\sigma > 1$. Now using the functional equation we get that the zeros of $\zeta(s)$ for $\sigma < 0$ are exactly the poles of $\Gamma(s/2)$ which gives that $\zeta(-2n) = 0$ for all integral $n \geq 1$. These are called the *trivial zeros* of $\zeta(s)$. All the other zeros must lie in the *critical strip* $0 \leq \sigma \leq 1$. We have the famous Riemann Hypothesis :

Conjecture 1.1. (Riemann Hypothesis) *All the non-trivial zeros of $\zeta(s)$ occur on the line $\sigma = 1/2$.*

It was Riemann who first realized the connection between distribution of zeros of $\zeta(s)$ and the distribution of prime numbers in $\mathbb{Z}$. However, it was not until 1896, that J. Hadamard and de la Vallée Poussin independently proved the prime number theorem from the fact that $\zeta(1 + it) \neq 0$, $\forall t \in \mathbb{R}$. We record this in the following two theorems (see [25] for details).

Theorem 1.2. *For all real t we have $\zeta(1 + it) \neq 0$.*

Theorem 1.3. *For $x \geq 0$, let $\pi(x)$ denote the number of prime numbers p in $\mathbb{Z}$ not greater than x . Then*

$$\pi(x) = \frac{x}{\log x} + o\left(\frac{x}{\log x}\right), \quad \text{as } x \rightarrow \infty$$

1.2. Dedekind ζ -function. Recall that a number field is a finite degree extension of the rational number field $\mathbb{Q}$. There is a generalization of the Riemann ζ -function, due to Dedekind, which associates to any number field K a function $\zeta_K(s)$ of a complex variable s .

Let K be a number field and let $\mathcal{O}_K$ be its ring of integers. By definition $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}$ in K . It is a fundamental fact that $\mathcal{O}_K$ is a Dedekind domain. (See Chapter 10 of [10].) For every (integral) ideal $\mathfrak{a}$ in $\mathcal{O}_K$ let $N(\mathfrak{a})$ denote the absolute norm of $\mathfrak{a}$, i.e. $N(\mathfrak{a}) = \#(\mathcal{O}_K/\mathfrak{a})$. The Dedekind ζ -function associated to K is given by :

$$(2) \quad \zeta_K(s) = \sum_{\mathfrak{a}} \frac{1}{N(\mathfrak{a})^s} ; \quad \text{for } \sigma > 1.$$

Note that $\zeta_{\mathbb{Q}}(s)$ is nothing but the Riemann zeta function considered in Section 1.1. The Dedekind zeta function has properties similar to the Riemann zeta function and these are summarized in the following theorem. (See Section 5.2 of [28].)

Theorem 1.4. 1. **Convergence :** *The series defining $\zeta_K(s)$ converges absolutely and uniformly on any compact subset of region $\sigma \geq 1 + \epsilon$ for any $\epsilon > 0$.*

2. **Euler Product :** *For $\sigma > 1$ we have*

$$\zeta_K(s) = \prod_{\mathfrak{p}} \left(1 - \frac{1}{N(\mathfrak{p})^s}\right)^{-1}$$

where the product runs over all the prime ideals in $\mathcal{O}_K$.

3. **Analytic continuation & Functional equation :** Let N be the degree of K over $\mathbb{Q}$. The function $\zeta_K(s)$ admits a meromorphic continuation to $\sigma > 1 - 1/N$ and is holomorphic except for a simple pole at $s = 1$ with residue given by :

$$\lim_{s \rightarrow 1^+} (s - 1)\zeta_K(s) = \frac{2^{r_1}(2\pi)^{r_2}Rh}{m|D|^{1/2}}$$

where h is the class number of K , r_1 is the number of real embeddings of K , r_2 is the number of complex embeddings of K , R is the regulator, m is the number of roots of unity and D is the discriminant of K over $\mathbb{Q}$. Thus $(s - 1)\zeta_K(s)$ extends to an entire function of s . Further let

$$\Phi_K(s) = (\pi^{-s/2}\Gamma(s/2))^{r_1}((2\pi)^{-s}\Gamma(s))^{r_2}|D|^{s/2}\zeta_K(s).$$

Then $\Phi_K(s)$ admits a meromorphic continuation to the entire s -plane and is holomorphic except for simple poles at $s = 0, 1$ and satisfies the functional equation :

$$\Phi_K(s) = \Phi_K(1 - s).$$

The residue formula is called the *Analytic class number formula* and is a deep formula. The reader may refer to Section 5.2 of [28] for some related matters.

From the Euler product it follows that $\zeta_K(s) \neq 0$ for $\sigma > 1$. Now one may use the functional equation and deduce the location of zeros of ζ_K using the information about poles of $\Gamma(s)$ for $\sigma < 0$ (which is well understood) and by knowing r_1 and r_2 . These zeros are called the *trivial zeros* of $\zeta_K(s)$. Notice that $\zeta_K(-n) = 0$ for all integers $n \geq 0$ if $r_2 \geq 1$. There is a ‘‘Riemann Hypothesis’’ for ζ_K which predicts the location of the non-trivial zeros. This is given below.

Conjecture 1.2. (Generalized Riemann Hypothesis) *All the non-trivial zeros of $\zeta_K(s)$ occur on the critical line $\sigma = 1/2$.*

The distribution of zeros of $\zeta_K(s)$ is connected with the distribution of prime ideals in $\mathcal{O}_K$. The Hadamard-de la Vallée Poussin theorem is valid as well for the Dedekind zeta function and this is used to prove the prime number theorem for number fields. These are recorded in the following two theorems.

Theorem 1.5. *For all real t we have $\zeta_K(1 + it) \neq 0$.*

Theorem 1.6. *Let $\pi_K(x)$ denote the number of prime ideals $\mathfrak{p}$ in $\mathcal{O}_K$ whose absolute norm $N(\mathfrak{p})$ is not greater than x . Then*

$$\pi_K(x) = \frac{x}{\log x} + o\left(\frac{x}{\log x}\right), \quad \text{as } x \rightarrow \infty.$$

1.3. Classical Dirichlet L -functions. Let N be an integer. A *Dirichlet character modulo N* is a character χ of $(\mathbb{Z}/N\mathbb{Z})^\times$ and is thought of as a function, also denoted χ , of $\mathbb{Z}$ by defining it to be 0 on integers which are not relatively prime to N . If $N = 1$ then this function, denoted $\mathbf{1}$, is the trivial Dirichlet character and takes the value 1 on every non-zero integer and vanishes on 0.

If M divides N then we have a canonical map $(\mathbb{Z}/N\mathbb{Z})^\times \rightarrow (\mathbb{Z}/M\mathbb{Z})^\times$. A Dirichlet character modulo N is said to be *primitive* if it cannot be factored via such a map for any proper divisor M of N . Given any Dirichlet character χ modulo N there exists an integer f_χ which divides N and χ factors through $(\mathbb{Z}/f_\chi\mathbb{Z})^\times$ and is primitive as a Dirichlet character modulo f_χ . This f_χ is called the *conductor of χ* . We shall henceforth deal only with primitive Dirichlet characters.

Let χ be a primitive Dirichlet character modulo N . To χ is attached a function of a complex variable s , called the *classical Dirichlet L -function* attached to χ and given by

$$(3) \quad L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} ; \quad \text{for } \sigma > 1.$$

Remark 1.1. Note that the classical Dirichlet L -function for the trivial Dirichlet character $\mathbf{1}$ is just the Riemann ζ -function dealt with in Section 1.1. So for the rest of this section we deal only with non-trivial Dirichlet characters.

Before we give the main theorem we need to introduce some Gauss sums. To a primitive Dirichlet character χ of modulus N is attached the *Gauss sum* $\tau(\chi)$ which is given by

$$(4) \quad \tau(\chi) = \sum_{m \bmod N} \chi(m) e^{2\pi i m/N}$$

The main fact we need about Gaussian sums is the easy to verify fact that $|\tau(\chi)|^2 = N$. Let w_χ be the complex number of absolute value 1 such that $\tau(\chi) = w_\chi \sqrt{N}$.

We now summarize the main properties of the classical Dirichlet L -functions in the following theorem. (See Section 1.1 of [7].)

Theorem 1.7. *Let χ be a primitive Dirichlet character modulo N . Assume that $N > 1$. Choose $\epsilon = 0, 1$ according to $\chi(-1) = (-1)^\epsilon$.*

1. **Convergence :** *The L -function $L(s, \chi)$ converges absolutely and uniformly on compact subsets of the region $\sigma > 1$.*

2. **Euler Product** : For $\sigma > 1$ we have

$$L(s, \chi) = \prod_p \left(1 - \frac{\chi(p)}{p^s} \right)^{-1}$$

where p runs over all the prime numbers in $\mathbb{Z}$.

3. **Analytic continuation & Functional equation** : Let

$$\Phi(s, \chi) = N^{s/2} \pi^{-s/2} \Gamma\left(\frac{s+\epsilon}{2}\right) L(s, \chi).$$

Since $\chi \neq \mathbf{1}$, one has that $\Phi(s, \chi)$ admits an analytic continuation to an entire function. In particular, if $\chi \neq \mathbf{1}$, then $L(s, \chi)$ extends to an entire function of s . Further it satisfies the following functional equation

$$\Phi(s, \chi) = (-i)^\epsilon w_\chi \Phi(1-s, \bar{\chi}).$$

In the previous sections we had seen that the non-vanishing of L -functions on the line $\sigma = 1$ had some arithmetic significance. The same is true for the classical Dirichlet L -function and gives the famous theorem of Dirichlet on primes in arithmetic progression. We need a notion of density to give an even precise statement on primes in arithmetic progression.

Definition 1.1. Let $\mathcal{S}$ be a set of primes in a number field K . The limit

$$\lim_{s \rightarrow 1^+} \frac{\sum_{\mathfrak{p} \in \mathcal{S}} N(\mathfrak{p})^{-s}}{-\log(s-1)}$$

if it exists, is called the **Dirichlet density** of $\mathcal{S}$.

Theorem 1.8. Let $N > 1$. Let χ be a primitive Dirichlet character modulo N . Then for all $t \in \mathbb{R}$ we have

$$L(1+it, \chi) \neq 0.$$

Theorem 1.9. Any arithmetic progression of the form $\{a + nd\}_{n \in \mathbb{N}}$ with the first term a being relatively prime to the constant difference d contains infinitely many primes. In fact the Dirichlet density of this set of primes is $\frac{1}{\phi(d)}$ where ϕ is the Euler ϕ -function.

1.4. Hecke L -functions. Let K be a number field and let $\mathfrak{m}$ be an integral ideal of K . Let $J_K^\mathfrak{m}$ be the group of fractional ideals of K which are relatively prime to $\mathfrak{m}$. Let $P_{K,+}^\mathfrak{m}$ be the subgroup of $J_K^\mathfrak{m}$ containing all those principal ideals (x) , for $x \in K^*$ for which x is totally positive and $x-1 \in \mathfrak{m}$. Then $J_K^\mathfrak{m}/P_{K,+}^\mathfrak{m}$ is finite.

Let ψ be a character of $J_K^\mathfrak{m}/P_{K,+}^\mathfrak{m}$. Such a character is called a *Dirichlet character of K modulo $\mathfrak{m}$* . As in [14], such a character is obtained by

a idele class character whose infinite components are discrete. Let $h_{\mathfrak{m}}$ denote the number of distinct Dirichlet characters of K modulo $\mathfrak{m}$. The character χ can be extended to a multiplicative map of J_K - the group of fractional ideals of K - by defining it to be 0 on all those ideals which are not relatively prime to $\mathfrak{m}$.

Let ψ be a Dirichlet character of K modulo $\mathfrak{m}$. To ψ is attached an L -function called the *Hecke L -function* and is given by

$$(5) \quad L(s, \psi) = \sum_{\mathfrak{a}} \frac{\psi(\mathfrak{a})}{N(\mathfrak{a})^s} ; \quad \text{for } \sigma > 1$$

where $\mathfrak{a}$ runs over all the integral ideals in J_K and for any integral ideal $\mathfrak{a}$, $N(\mathfrak{a})$ denotes the absolute norm of $\mathfrak{a}$, which is the cardinality of $\mathcal{O}_K/\mathfrak{a}$.

If we take $\mathfrak{m} = (1)$ and ψ the trivial character $\mathbf{1}$ then we have $L(s, \mathbf{1}) = \zeta_K(s)$. If we take $K = \mathbb{Q}$ and $\mathfrak{m} = (m)$ for some integer m then it is easy to see that $J_K^{\mathfrak{m}}/P_{K,+}^{\mathfrak{m}} = (\mathbb{Z}/m\mathbb{Z})^{\times}$. And so we also get the classical Dirichlet L -functions as particular cases of Hecke L -functions.

We remark here that what we call Hecke L -functions are in some literature called Dirichlet L -functions, for example in [14] and [28]. Hecke himself considered slightly more general characters which he called grossencharacters. To such a grossencharacter is attached an L -function and in this setting he proved the analytic continuation and functional equation for such functions. This gives, in the particular case when a grossencharacter is a Dirichlet character of K modulo $\mathfrak{m}$, the following theorem.

Theorem 1.10. *Let K be a number field of degree N over $\mathbb{Q}$. Let $\mathfrak{m}$ be an integral ideal of K . Let ψ be a Dirichlet character of K modulo $\mathfrak{m}$. Then :*

1. **Convergence :** *The Hecke L -function converges for $\sigma > 1 - 1/N$. It converges absolutely and uniformly on compact subsets of $\sigma > 1$.*
2. **Euler Product :** *For $\sigma > 1$ we have*

$$L(s, \psi) = \prod_{\mathfrak{p}} \left(1 - \frac{\psi(\mathfrak{p})}{N(\mathfrak{p})^s} \right)^{-1}$$

where $\mathfrak{p}$ runs over all the prime ideals of K and $N(\mathfrak{p})$ denotes the absolute norm of $\mathfrak{p}$.

3. **Analytic continuation & Functional equation :** *Let*

$$\Phi(s, \psi) = \prod_{i=1}^{r_1} \Gamma\left(\frac{s + a_i}{2}\right) \Gamma(s)^{r_2} \left(\frac{|D_K| N(\mathfrak{f}_{\psi})}{4^{r_2} \pi^N}\right)^{s/2} L(s, \psi)$$

where r_1 and r_2 are respectively the number of real and complex valuations on K , D_K is the discriminant of K , $\mathfrak{f}_\psi$ is the conductor of ψ (refer [14] for a definition of conductor of ψ) and a_i is 0 or 1 according as $\psi((x))$ depends or not on the sign of the i -th real embedding of x for any x such that $x - 1 \in \mathfrak{m}$.

Then $\Phi(s, \psi)$ extends to a meromorphic function of the entire s -plane and there exists a constant $W = W(\psi)$ of absolute value 1 such that

$$\Phi(s, \psi) = W(\psi)\Phi(1 - s, \overline{\psi}).$$

It follows from the proof of the functional equation that $L(s, \psi)$ extends to an entire function of s if ψ is not the trivial character and if ψ is trivial then $L(s, \psi)$ is holomorphic everywhere except for a simple pole at $s = 1$.

As in the ‘classical’ number theory of Sections 1.1, 1.2 and 1.3 the Hecke L -functions can be used to derive the following ‘arithmetic theorems.’ Refer to [14] for a proof of all the following theorems in this section.

Theorem 1.11. (Hadamard - de la Vallee Poussin - Landau) For all $t \in \mathbb{R}$ we have $L(1 + it, \psi) \neq 0$.

Theorem 1.12. (The prime ideal theorem) If ψ is not the trivial character then

$$\sum_{N(\mathfrak{p}) \leq x} \psi(\mathfrak{p}) = o\left(\frac{x}{\log x}\right), \quad \text{as } x \rightarrow \infty.$$

If ψ is the trivial character then it boils down to Theorem 1.6 since the Hecke L -function is then the Dedekind zeta function $\zeta_K(s)$.

2. ARTIN L -FUNCTIONS

2.1. Definition. Let E/F be a Galois extension of number fields and let $G = \text{Gal}(E/F)$ be the Galois group. Let (ϕ, V) be a representation of G on a finite dimensional complex vector space V . An Artin L -function is a meromorphic function, denoted $L(s, \phi, E/F)$, attached to this data.

We now collect some useful number theoretic information on prime ideals in number fields and their ramification theory in extensions of number fields.

Let E/F be an extension of number fields. Let $\mathfrak{p}$ be a prime ideal in $\mathcal{O}_F$. Let $\mathfrak{p}\mathcal{O}_E = \mathcal{P}_1^{e_1} \cdots \mathcal{P}_g^{e_g}$ be the factorization of $\mathfrak{p}\mathcal{O}_E$ into prime ideals in $\mathcal{O}_E$. (This is possible as the ring of integers of a number field is a Dedekind domain.) Then $\{\mathcal{P}_1, \dots, \mathcal{P}_g\}$ are all the prime ideals of

$\mathcal{O}_E$ which lie above $\mathfrak{p}$. Since E/F is Galois, the Galois group G acts transitively on the set $\{\mathcal{P}_1, \dots, \mathcal{P}_g\}$. Let $\mathcal{P}$ be one of the primes $\mathcal{P}_i$. Let

$$D_{\mathcal{P}} := \{\tau \in G : \tau^{\tau} = \mathcal{P}\}.$$

This is called the *decomposition group* of $\mathcal{P}$. Let

$$I_{\mathcal{P}} := \{\tau \in G : \tau(x) \equiv x \pmod{\mathcal{P}} \quad \forall x \in \mathcal{O}_E\}.$$

This is called the *inertia group* of $\mathcal{P}$. We will also need a filtration of subgroups of the decomposition group given by congruence conditions. We will let $G_0(\mathcal{P})$ denote the decomposition group $D_{\mathcal{P}}$ and for all $i \geq 1$ denote

$$G_i(\mathcal{P}) := \{\tau \in G : \tau(x) \equiv x \pmod{\mathcal{P}^i} \quad \forall x \in \mathcal{O}_E\}.$$

If there is no confusion about the prime $\mathcal{P}$ then G_i will denote $G_i(\mathcal{P})$.

Let $\overline{E}_{\mathcal{P}}$ and $\overline{F}_{\mathfrak{p}}$ denote the residue fields $\mathcal{O}_E/\mathcal{P}$ and $\mathcal{O}_F/\mathfrak{p}$ respectively. There is an isomorphism $D_{\mathcal{P}}/I_{\mathcal{P}} \simeq \text{Gal}(\overline{E}_{\mathcal{P}}/\overline{F}_{\mathfrak{p}})$. Now $\overline{E}_{\mathcal{P}}/\overline{F}_{\mathfrak{p}}$ is a Galois extension of finite fields and hence $D_{\mathcal{P}}/I_{\mathcal{P}}$ is a finite cyclic group generated by $\sigma_{\mathcal{P}} \in D_{\mathcal{P}}/I_{\mathcal{P}}$ which is (and reasonably so) called the *Frobenius automorphism* and is characterised by

$$\sigma_{\mathcal{P}}(x) \equiv x^{N(\mathfrak{p})} \pmod{\mathcal{P}} \quad \forall x \in \mathcal{O}_E.$$

Here $N(\mathfrak{p})$ denotes the absolute norm of $\mathfrak{p}$ which is just the cardinality of the finite field $\overline{F}_{\mathfrak{p}}$. Note that the Frobenius automorphism is well-defined only modulo the inertia.

The prime $\mathfrak{p}$ is unramified in E/F if and only if $I_{\mathcal{P}} = \{1\}$ for any prime $\mathcal{P}$ above $\mathfrak{p}$. Similarly the prime $\mathfrak{p}$ splits completely in E/F if and only if $D_{\mathcal{P}} = I_{\mathcal{P}} = \{1\}$ for any prime $\mathcal{P}$ above $\mathfrak{p}$.

Since the action of the Galois group on the set of primes in E above $\mathfrak{p}$ is transitive we get that for two primes $\mathcal{P}_1$ and $\mathcal{P}_2$ above $\mathfrak{p}$ there exists an element in G which simultaneously conjugates $D_{\mathcal{P}_1}$ into $D_{\mathcal{P}_2}$, $I_{\mathcal{P}_1}$ into $I_{\mathcal{P}_2}$ and $\sigma_{\mathcal{P}_1}$ into $\sigma_{\mathcal{P}_2}$. This implies the following :

$$\det(1 - N(\mathfrak{p})^{-s} \phi(\sigma_{\mathcal{P}_1})|_{V^{I_{\mathcal{P}_1}}}) = \det(1 - N(\mathfrak{p})^{-s} \phi(\sigma_{\mathcal{P}_2})|_{V^{I_{\mathcal{P}_2}}})$$

We are now in a position to define an Artin L -function. This is a function attached to a finite dimensional representation (ϕ, V) of G given by

$$(6) \quad L(s, \phi, E/F) = \prod_{\mathfrak{p}} \det(1 - N(\mathfrak{p})^{-s} \phi(\sigma_{\mathcal{P}})|_{V^{I_{\mathcal{P}}}})^{-1}$$

where $\mathfrak{p}$ runs over all the prime ideals in $\mathcal{O}_F$.

It is also clear that $\det(1 - N(\mathfrak{p})^{-s} \phi(\sigma_{\mathcal{P}})|_{V^{I_{\mathcal{P}}}})$ is independent of the choice of $\sigma_{\mathcal{P}}$ in the conjugacy class (in G) containing $\sigma_{\mathcal{P}}$. So we may replace ϕ by its character $\chi = \chi_{\phi}$. So henceforth we are going to talk in

terms of characters of representations and also change notation and we let $L(s, \chi, E/F)$ denote the Artin L -function attached to a character χ of G .

We will now give what is called the *completed* Artin L -function. (The reader may refer to [18] for details on completed Artin L -functions.) These completed functions play a crucial role in describing the functional equation satisfied by an Artin L -function. We will not really be using these completed L -functions for the rest of the course. The reason it is included here is so that the reader may see the functional equation satisfied by Artin L -functions which will be helpful in Section 4.

Let

$$\Lambda(s, \chi) := A(\chi)^{s/2} \gamma(s, \chi) L(s, \chi, E/F)$$

where $A(\chi)$ is a constant and $\gamma(s, \chi)$ is composed of Gamma functions as described below. Let $\gamma(s) = \pi^{-s/2} \Gamma(s/2)$ and let

$$\gamma(s, \chi) = \prod_{\mathfrak{p}|\infty} \gamma_{\mathfrak{p}}(s, \chi)$$

where

$$\gamma_{\mathfrak{p}}(s, \chi) = \begin{cases} (\gamma(s) \gamma(s+1))^n & , \text{ if } \mathfrak{p} \text{ is complex} \\ \gamma(s)^{n_{\mathfrak{p}}^+} \gamma(s+1)^{n_{\mathfrak{p}}^-} & , \text{ if } \mathfrak{p} \text{ is real,} \end{cases}$$

the number n is the degree of χ , i.e. $n = \chi(1)$. If $\mathfrak{p}$ is real then let $\mathcal{P}$ be a place in E above $\mathfrak{p}$. Then the decomposition group $D_{\mathcal{P}}$ is of order 1 or 2 and let it be generated by w . The integer $n_{\mathfrak{p}}^+$ is the dimension of the subspace of V fixed by $\phi(w)$ and the integer $n_{\mathfrak{p}}^-$ is the codimension of the same subspace.

The constant $A(\chi)$ is given by the formula

$$A(\chi) = |D_F|^n N(\mathfrak{f}(\chi))$$

where $|D_F|$ is the discriminant of $F/\mathbb{Q}$ and $\mathfrak{f}(\chi)$ is the so-called *Artin Conductor* which is defined by :

$$\mathfrak{f}(\chi) := \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}(\chi)}$$

the product being over all the prime ideals in $\mathcal{O}_F$. The quantity $n_{\mathfrak{p}}(\chi)$ is given by

$$n_{\mathfrak{p}}(\chi) := \sum_i (|G_i|/|G_0|) \text{codim}_V(V^{G_i})$$

and it is a theorem due to Artin that $n_{\mathfrak{p}}(\chi)$ is indeed an integer and is zero for almost all primes. (See [33] for a proof of this theorem.)

Theorem 2.1. *Let G be the Galois group of a Galois extension E/F of number fields. Let χ be an irreducible character of G . The completed Artin L -function $\Lambda(s, \chi)$ admits a meromorphic continuation to the*

entire s -plane and there exists a constant $W = W(\chi)$ of absolute value 1 such that

$$\Lambda(s, \chi) = W(\chi)\Lambda(1 - s, \bar{\chi}).$$

The constant $W = W(\chi)$ which occurs in the theorem is called the *Artin root number*. This theorem is due to Brauer and the main tool in its proof is the Brauer's induction formula. Refer to [18] for a proof of this theorem and also for further properties of the root number.

2.2. Some easy properties. In this section we list some basic properties about Artin L -functions. References for the theorems in this section are [14], [18] and Chapter 5 of [28]. The first theorem, due to Artin, describes the dependence of the L -function on the character χ .

Theorem 2.2. *Let G be the Galois group of a Galois extension E/F of number fields. Let H and N be subgroups of G and assume that N is normal in G . Let χ, χ_1 and χ_2 be characters of G . Let ψ be a character of H . Let $\text{Ind}_H^G(\psi)$ denote the character of G obtained by inducing ψ from H to G . (Sometimes we denote this induced character simply by ψ^G .) Let $\bar{\tau}$ be a character of G/N and let τ be the pull back of $\bar{\tau}$ to G . Then :*

1. $L(s, \chi_1 \oplus \chi_2, E/F) = L(s, \chi_1, E/F)L(s, \chi_2, E/F)$.
2. $L(s, \psi, E/E^H) = L(s, \text{Ind}_H^G(\psi), E/F)$.
3. $L(s, \bar{\tau}, E^N/F) = L(s, \tau, E/F)$.

The next theorem, also due to Artin, gives the relation between Artin L -functions and the Dedekind zeta functions associated to number fields. Statement (3) below, which gives a factorization of the Dedekind zeta function of a number field relative to a field it contains, is called the *Artin-Takagi factorization*.

Theorem 2.3. *Let G be the Galois group of a Galois extension E/F of number fields. Let $\mathbf{1}_G$ be the trivial character of G . Let $\mathbf{R}_G$ be the regular character of G . Then :*

1. $L(s, \mathbf{1}_G, E/F) = \zeta_F(s)$
2. $L(s, \mathbf{R}_G, E/F) = \zeta_E(s)$
3. $\zeta_E(s) = \prod_{\chi \in \hat{G}} L(s, \chi, E/F)^{\chi(1)}$, where $\hat{G}$ denotes the set of irreducible characters of G .

Proof : Statement (1) follows from Statement (3) of Theorem ???. Statement (2) follows from Statement (1) and Statement (2) of Theorem 2.2. Statement (3) follows from Statement (2). $\square$

2.3. Artin reciprocity law. In this section we give a brief summary of those essential definitions and statements from global class field theory which lead to the celebrated Artin reciprocity law. The summary is based on the presentation in [28].

Let K be a number field of degree N over $\mathbb{Q}$. Let A_K be the adèle ring of K . Let I_K be the group of ideles of K and let $C_K = I_K/K^*$ be the idele class group of K .

Let $\mathfrak{m}$ be a cycle of K by which is meant a formal product $\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$ where $n_{\mathfrak{p}} \geq 0$ for all $\mathfrak{p}$. Further almost all the $n_{\mathfrak{p}}$ are zero and if $\mathfrak{p}$ is Archimedean then $n_{\mathfrak{p}}$ is 0 or 1. If p is finite, we set $U_p^{n_p}$ to be the group of n_p -th principal units of K (where $U_p^0 = U_p$). If p is complex, we set $U_p^{n_p}$ to be $\mathbb{C}^*$. If p is real, we set $U_p^{n_p}$ to be $\mathbb{R}_+$ or $\mathbb{R}^*$ according as $n_p = 1$ or 0.

To a cycle $\mathfrak{m}$ associate a subgroup $I_K^{\mathfrak{m}}$ of I_K consisting of all ideles α such that

- (1) If $\mathfrak{p}$ is a finite prime then $\alpha_{\mathfrak{p}} \in U_{\mathfrak{p}}^{n_{\mathfrak{p}}}$.
- (2) If $\mathfrak{p}$ is a real place and $n_{\mathfrak{p}} = 1$ then $\alpha_{\mathfrak{p}} > 0$. (These conditions can be said to be expressed as $\alpha \equiv 1(\mathfrak{m})$.)

This gives rise to a *congruence subgroup modulo $\mathfrak{m}$* , denoted $C_K^{\mathfrak{m}}$, defined as $C_K^{\mathfrak{m}} = I_K^{\mathfrak{m}} \cdot K^*/K^*$. The factor group $C_K/C_K^{\mathfrak{m}}$ is called the *idele theoretic ray class group of K modulo $\mathfrak{m}$* .

Theorem 2.4. (Existence theorem) *The map $L \mapsto \mathcal{N}_L = N_{L/K}(C_L)$ gives a bijection between abelian extensions of K and closed subgroups of finite index in C_K . This bijection is inclusion reversing and preserves the obvious lattice structures on both sides.*

The closed subgroups of finite index of C_K are called *norm groups* of C_K . If $\mathcal{N}$ is a norm group of C_K then the abelian extension L/K realizing $\mathcal{N}$ is called the *class field of $\mathcal{N}$* .

Theorem 2.5. *The norm groups of C_K are precisely the subgroups of C_K which contain a congruence subgroup $C_K^{\mathfrak{m}}$ for some cycle $\mathfrak{m}$.*

Let $\mathfrak{m}$ be a cycle of K . The class field $K^{\mathfrak{m}}$ of the congruence subgroup $C_K^{\mathfrak{m}}$ is called the *ray class field of K modulo $\mathfrak{m}$* .

If $K = \mathbb{Q}$ and $\mathfrak{m} = p_{\infty}m$ where p_{∞} is the infinite prime of $\mathbb{Q}$ and m is some integer, then $\mathbb{Q}^{\mathfrak{m}} = \mathbb{Q}(\mu_m)$ where μ_m is the group of m -th roots of unity in $\mathbb{C}$. So the ray class fields of a number field are a generalization of cyclotomic extensions of $\mathbb{Q}$ and Theorems 2.4 and 2.5 give the general form of the famous Kronecker-Weber theorem.

If L/K is an abelian extension then a cycle $\mathfrak{m}$ of K is called a *cycle of declaration for L* , if $L \subset K^{\mathfrak{m}}$.

One can talk about the idele theoretic ray class groups of a number field in terms of ideals. Let $J_K^{\mathfrak{m}}$ be the group of fractional ideals of K which are relatively prime to $\mathfrak{m}$. Let $P_K^{\mathfrak{m}}$ be the group of principal fractional ideals (x) with $x \equiv 1(\mathfrak{m})$. The group $J_K^{\mathfrak{m}}/P_K^{\mathfrak{m}}$ is called the *ideal theoretic ray class group of K modulo $\mathfrak{m}$* and it is always a finite group. If $\mathfrak{m} = (1)$ then it is the usual class group of K . The following lemma justifies the terminology.

Lemma 2.1. *Let K be a number field and let $\mathfrak{m}$ be a cycle of K . Then $C_K/C_K^{\mathfrak{m}}$ is canonically isomorphic to $J_K^{\mathfrak{m}}/P_K^{\mathfrak{m}}$.*

We are now in a position to state the Artin reciprocity law.

Theorem 2.6. (Artin reciprocity law) *Let L/K be an abelian extension of number fields. Let $\mathfrak{m}$ be a cycle of declaration for L/K . There is a surjective homomorphism (called the Artin Symbol) from $J_K^{\mathfrak{m}}/P_K^{\mathfrak{m}}$ onto the Galois group $\text{Gal}(L/K)$ whose kernel is $H_{L/K}^{\mathfrak{m}}/P_K^{\mathfrak{m}}$ where $H_{L/K}^{\mathfrak{m}} = N_{L/K}(J_L^{\mathfrak{m}})P_K^{\mathfrak{m}}$.*

Further if χ is a character of $\text{Gal}(L/K)$ and corresponds via the Artin symbol to a Dirichlet character ψ then the respective Artin L -function and the Hecke L -function are equal, i.e. $L(s, \chi, L/K) = L(s, \psi)$.

Corollary 2.1. *Let G be the Galois group of a Galois extension E/F of number fields. Let χ be a monomial character of G , i.e. χ is induced from a one dimensional character of some subgroup of G . Then the Artin L -function $L(s, \chi, E/F)$ extends to an entire function of s if χ does not contain the trivial character.*

Proof : Follows from Theorem 2.6 and Theorem 1.10. □

2.4. The main conjectures. In this section we state the basic conjectures regarding Artin L -functions. We start with the Dedekind conjecture.

Conjecture 2.1. (Dedekind) *Let E/F be an extension of number fields. Then the quotient $\zeta_E(s)/\zeta_F(s)$ extends to an entire function of s .*

Some progress has been made on the Dedekind conjecture. There are two notable theorems. The first, due independently to Brauer and Aramata, gives an affirmative answer when the extension E/F is a Galois extension. A simple proof of Brauer-Aramata theorem (cf. Corollary 3.1) is presented later on. The second theorem, due independently to Uchida [38] and van der Waall [40], gives an affirmative answer when the extension is a sub-extension of a solvable extension.

Murty and Raghuram [27] have a generalization of the Uchida-van der Waall theorem in terms of Artin L -functions.

We now state the Artin conjecture. This is still open and this course of lectures revolves around this conjecture and reflects on some programs currently in progress which hope to settle it.

Conjecture 2.2. (Artin) *Let G be the Galois group of a Galois extension E/F of number fields. Let χ be an irreducible character of G which is not the trivial character. Then $L(s, \chi, E/F)$ extends to an entire function of s .*

The following proposition says that it is enough to prove the Artin conjecture for Galois extensions of $\mathbb{Q}$.

Proposition 2.1. *If for all irreducible non-trivial characters χ of G , the Galois group of a Galois extension $K/\mathbb{Q}$, the Artin L -function $L(s, \chi, K/\mathbb{Q})$ extends to an entire function of s then the Artin conjecture is true.*

Proof : Let E/F be a Galois extension of number fields. Let $K/\mathbb{Q}$ be the Galois closure of $E/\mathbb{Q}$. Let G be the Galois group of $K/\mathbb{Q}$ and let H be the Galois group of E/F . Then H is a quotient of a subgroup of G . Let τ be any irreducible non-trivial character of H . Inflate τ to a character of the Galois group of K/F . Let θ be the character of G obtained by inducing τ from $\text{Gal}(K/F)$ to G . By Frobenius reciprocity θ does not contain the trivial character. By Hypothesis and Statement (1) of Theorem 2.2, $L(s, \theta, K/\mathbb{Q})$ is entire. But then by Statements (2) and (3) of Theorem 2.2 we have $L(s, \theta, K/\mathbb{Q}) = L(s, \tau, E/F)$. $\square$

There is a related conjecture which is in the spirit of the Artin's reciprocity Theorem 2.6. This is part of a vast program called the Langlands program. This conjecture, in the literature, is also called *The strong Artin conjecture* or the *Langlands conjecture* or the *Reciprocity conjecture*.

Conjecture 2.3. (Langlands) *Let G be the Galois group of a Galois extension E/F of number fields. Let (ϕ, V) be an n dimensional irreducible representation of G . Then there exists a cuspidal automorphic representation π of $GL(n, \mathbb{A}_F)$ such that*

$$L(s, \phi, E/F) = L(s, \pi).$$

Here $L(s, \pi)$ is the L -function attached by Langlands to the representation π .

See Sections 3.3 and 3.9 of [7] for the definition of a cuspidal automorphic representation of $GL(n, \mathbb{A}_F)$ and the associated L -function. See [5] for a definitive and a very general account of Langlands L -functions.

Remark 2.1. By the work of Godement and Jacquet [13] it is known that $L(s, \pi)$ is entire for a cuspidal automorphic representation π of $GL(n, \mathbb{A}_F)$. So if the strong Artin conjecture is true then so is the Artin conjecture. However there are cases where the Artin conjecture is known and the Langlands conjecture is not known. For example, when G is supersolvable, we know Artin's conjecture but not the stronger Artin conjecture. In this case, it can be shown that every irreducible character is monomial (See [33] for a proof of this theorem.) So the Artin conjecture follows in the supersolvable case by using Artin reciprocity.

A cuspidal automorphic representation of $GL(1, \mathbb{A}_F)$ is nothing but a Hecke character of F . (The characters discussed in section 1.4 are a special case.) Hence the Langlands conjecture is a direct generalization of the Artin reciprocity law in the case $n = 1$. That is why the Langlands program is considered as a search for a “non-abelian class field theory”.

We now give a brief survey of the major theorems which throw light on the Artin conjecture. The first of these is already mentioned in Corollary 2.1. We mention this again below.

Theorem 2.7. (Artin) *Let G be the Galois group of a Galois extension E/F of number fields. Let χ be a non-trivial one-dimensional character of G . Then $L(s, \chi, E/F)$ extends to an entire function of s .*

There has been throughout this century some progress made in representations of finite groups in an effort to find and prove theorems which might help in a successful resolution of the Artin conjecture. There is a fundamental theorem due to Brauer which says that any irreducible character of G is a $\mathbb{Z}$ -span of monomial characters. This combined with Artin reciprocity is the standard way of proving that an Artin L -function admits a meromorphic continuation to the entire s -plane. This theme is taken up in Section 3.1.

We now deal with the case of two dimensional representations of G . One knows by the work of Felix Klein, that all the finite subgroups of $PGL(2, \mathbb{C})$ are the following:

1. Cyclic group of any order.
2. Dihedral group of any order.
3. Tetrahedral group which is isomorphic to A_4 .
4. Octahedral group which is isomorphic to S_4 .

5. Icosahedral group which is isomorphic to A_5 .

This gives that any finite subgroup of $GL(2, \mathbb{C})$ is a (finite) cyclic central extension of a finite subgroup of $PGL(2, \mathbb{C})$.

Now from easy facts from the representations of finite groups and from Artin reciprocity the Artin conjecture follows in the cyclic and dihedral case (i.e. if (ϕ, V) is a representation of G on a two dimensional vector space then the image $\phi(G)$ in $PGL(V)$ is cyclic or dihedral).

In the tetrahedral case it is a non-trivial theorem of Langlands who used techniques of base change and proved the strong Artin conjecture. Later Tunnell extended the work of Langlands and proved the strong Artin conjecture in the octahedral case. These statements are summarized in the following theorem (see [17] and [37] for details).

Theorem 2.8. (Artin-Langlands-Tunnell) *Let G be the Galois group of a Galois extension E/F of number fields. Let (ϕ, V) be a two dimensional representation of G . Let $\phi(G)$ be a solvable subgroup of $GL(V)$. Then $L(s, \phi, E/F)$ extends to an entire function of s .*

There are presently four programs running and any one of them if successfully completed will give a proof of the Artin conjecture. These programs are the following :

1. *The Langlands Program* : This involves, in its full generality, heavy doses of harmonic analysis on reductive groups over local and global fields. The part of it we are interested in is already alluded to and is embodied in the strong Artin Conjecture 2.3. We shall not be going into this approach. See [2], [6], [12], [21], [29] and [30] for an indication regarding this program and its relation to the Artin conjecture.
2. *Serre's conjecture* : There is a conjecture of Serre in [34] which says that a two dimensional representation over a finite field of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which satisfies some conditions (odd and absolutely irreducible) is modular. We will not elaborate any further on this matter and refer the reader to [34] for further details. The connection between Serre's conjecture and Artin L -functions is a theorem due to Chandrashekhhar Khare (see Proposition 1 of [16]) that Serre's conjecture implies the Artin conjecture for two dimensional representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. We remark here that the Artin conjecture is stated differently in [16] but it coincides with the statement we have given by using the classical converse

theorem of Weil and Langlands. (See the discussion before Proposition 1 in [16].) For recent progress in this theme, we refer the reader to [8] and [36].

3. *Brauer-Heilbronn-Stark approach* : This involves representation theory of finite groups. Section 3 deals with this approach in some detail.
4. *Selberg's program* : This is in the realms of analytic number theory and will be dealt with in detail in Sections 4 and 5.

3. THE BRAUER - HEILBRONN - STARK APPROACH

3.1. Brauer's theorem. In this section we present analytic consequences of a famous theorem of Brauer which says that for a finite group G , its monomial characters generate over $\mathbb{Z}$ the Grothendieck group associated to G . This theorem may be considered as a prototypical example regarding statements in finite group theory which shed light on Artin L -functions. This theorem in particular gives a meromorphic continuation for any Artin L -function. In this section we shall come across more of such group theoretic theorems which have analytic implications.

Theorem 3.1. (Brauer) *Let G be a finite group. Let χ be an irreducible character of G . Then for some integer $r \geq 1$ and for $1 \leq i \leq r$ there exist one dimensional characters ψ_i of nilpotent subgroups H_i and there exist integers n_i such that*

$$\chi = \sum_{i=1}^r n_i \text{Ind}_{H_i}^G(\psi_i).$$

See [33] for a proof of the above theorem.

Theorem 3.2. (Brauer) *Let G be the Galois group of a finite Galois extension E/F of number fields. Let χ be an irreducible character of G . Then the Artin L -function $L(s, \chi, E/F)$ admits a meromorphic continuation to the entire s -plane.*

Proof : By Theorem 3.1 we have

$$L(s, \chi, E/F) = \prod_{i=1}^r L(s, \text{Ind}_{H_i}^G(\psi_i), E/F)^{n_i}.$$

By Theorem 2.2 we have $L(s, \text{Ind}_{H_i}^G(\psi_i), E/F) = L(s, \psi_i, E/E^{H_i})$ and this by Theorem 2.6 and by Theorem 1.10 extends to an entire function of s . Since the n_i 's are integers we get a meromorphic continuation of $L(s, \chi, E/F)$. $\square$

3.2. Heilbronn's character. We now study the behaviour of an Artin L -function at an arbitrary but fixed point $s_0 \in \mathbb{C}$.

Let G be the Galois group of a Galois extension E/F of number fields. Let χ be an irreducible character of G . Let the order of the Artin L -function $L(s, \chi, E/F)$ at $s = s_0$ be denoted by $n(G, \chi)$, i.e.

$$(7) \quad n(G, \chi) = \text{ord}_{s=s_0} L(s, \chi, E/F).$$

The **Heilbronn character** is not really a character but a class function defined on G given by the formula :

$$(8) \quad \Theta_G(g) = \sum_{\chi \in \widehat{G}} n(G, \chi) \chi(g).$$

If Artin's conjecture is true and $s_0 \neq 1$, then Θ_G is a character. A basic observation due to Heilbronn [15] is the following proposition.

Proposition 3.1. (Heilbronn) *Let G be the Galois group of a Galois extension E/F of number fields. Let H be a subgroup of G . Then*

$$\Theta_G|_H = \Theta_H.$$

Here Θ_H is the Heilbronn character for the extension E/E^H .

Proof : For the sake of brevity we write ψ^G for $\text{Ind}_H^G(\psi)$. We have using Frobenius reciprocity and orthogonality of characters the following equations :

$$\begin{aligned} \Theta_G|_H &= \sum_{\chi \in \widehat{G}} n(G, \chi) \chi|_H \\ &= \sum_{\chi \in \widehat{G}} n(G, \chi) \left(\sum_{\psi \in \widehat{H}} \langle \chi|_H, \psi \rangle_H \psi \right) \\ &= \sum_{\psi \in \widehat{H}} \left(\sum_{\chi \in \widehat{G}} n(G, \chi) \langle \chi, \psi^G \rangle_G \right) \psi. \end{aligned}$$

We may now use orthogonality and Theorem 2.2 as follows :

$$\begin{aligned} n(H, \psi) &= \text{ord}_{s=s_0} L(s, \psi, E/E^H) \\ &= \text{ord}_{s=s_0} L(s, \psi^G, E/F) \\ &= \text{ord}_{s=s_0} \prod_{\chi \in \widehat{G}} L(s, \chi, E/F)^{\langle \chi, \psi^G \rangle_G} \\ &= \sum_{\chi \in \widehat{G}} n(G, \chi) \langle \chi, \psi^G \rangle_G. \end{aligned}$$

The proposition follows. $\square$

3.3. A useful inequality. An easy application of finite group theory to the Heilbronn character gives an inequality involving the orders of Artin L -functions and the order of the Dedekind zeta function of the top field. This inequality has amazing consequences. We start with the theorem:

Theorem 3.3. [11] *Let G be the Galois group of a Galois extension E/F of number fields. Then*

$$\sum_{\chi \in \widehat{G}} n(G, \chi)^2 \leq (\text{ord}_{s=s_0} \zeta_E(s))^2$$

Proof : We have taking the inner product of Θ_G with itself,

$$\sum_{\chi \in \widehat{G}} n(G, \chi)^2 = \langle \Theta_G, \Theta_G \rangle_G = \frac{1}{|G|} \sum_{x \in G} |\Theta_G(x)|^2.$$

Let $\langle x \rangle$ denote the cyclic subgroup of G generated by an element x in G . We have using Proposition 3.1

$$\Theta_G(x) = \Theta_{\langle x \rangle}(x) = \sum_{\psi \in \widehat{\langle x \rangle}} n(\langle x \rangle, \psi) \psi(x).$$

By Theorem 2.6 we have $n(\langle x \rangle, \psi) \geq 0$. Hence we get

$$|\Theta_G(x)| \leq \sum_{\psi \in \widehat{\langle x \rangle}} n(\langle x \rangle, \psi).$$

Using the Artin-Takagi factorization (Theorem 2.3) we get

$$\sum_{\psi \in \widehat{\langle x \rangle}} n(\langle x \rangle, \psi) = \text{ord}_{s=s_0} \zeta_E(s).$$

Putting all these together gives us the theorem. $\square$

We now give the (seemingly endless !) corollaries to this theorem.

Corollary 3.1. (Brauer-Aramata) *The Dedekind conjecture is true for Galois extensions, i.e. if E/F is a Galois extension then $\zeta_E(s)/\zeta_F(s)$ is entire.*

Proof : Follows from Theorem 3.3 and the fact that $L(s, \mathbf{1}_G, E/F)$ is $\zeta_F(s)$. $\square$

Corollary 3.2. *Let G be the Galois group of a Galois extension of number fields E/F . Let χ be any irreducible character of G . Then for all $t \in \mathbb{R}$ we have*

$$L(1 + it, \chi, E/F) \neq 0.$$

Proof : Follows from Theorems 1.1, 1.2 and 3.3. $\square$

Corollary 3.3. (Stark) *Let G be the Galois group of a Galois extension of number fields E/F . If $\zeta_E(s)$ has a simple zero at $s = s_0$ then $L(s, \chi, E/F)$ is regular at $s = s_0$ for any irreducible character χ of G .*

Proof : The right hand side of the inequality in Theorem 3.3 is 1. So either the conclusion is valid or there is a unique irreducible χ_0 for which $n(G, \chi_0) = -1$. By Artin reciprocity, χ_0 is not one dimensional. This gives a contradiction by looking at the Artin-Takagi factorization. $\square$

Corollary 3.4. (Foote - Murty) *Let E/F be a solvable extension of number fields with Galois group G . Let $|G|$ be odd. If $\zeta_E(s)$ has a zero at $s = s_0$ of order at most 3 then $L(s, \chi, E/F)$ is regular at $s = s_0$ for any irreducible character χ of G .*

Proof : See [11] or [24]. This result is not an *easy* consequence of the above theorem though it begins its analysis from it. It involves rather intricate group theory for its proof. $\square$

Corollary 3.5. *Any pole of an Artin L -function $L(s, \chi, E/F)$, for χ irreducible, must occur among the zeros of $\zeta_E(s)$*

Proof : Obvious. $\square$

Corollary 3.6. *If the Dedekind zeta function $\zeta_E(s)$ satisfies the Riemann Hypothesis, i.e. if every non-trivial zero of $\zeta_E(s)$ is on the critical line $\sigma = 1/2$ then the Artin L -function $L(s, \chi, E/F)$, for χ irreducible, is regular in the half plane $\sigma > 1/2$.*

Proof : Under the Hypothesis, $\text{ord}_{s=s_0} \zeta_E(s) = 0$ for $\sigma_0 > 1/2$, the lemma follows from the inequality in Theorem 3.3. $\square$

3.4. Some variations. In this section we present some variations of Theorem 3.3. The idea being that since Artin L -functions of abelian characters are anyway entire, one should try to get an inequality as in Theorem 3.3 without the terms corresponding to the abelian characters. We look at the specific case when the Galois group is solvable. As a first step is the following theorem due to the author [24].

Theorem 3.4. *Let G be the Galois group of a solvable Galois extension E/F of number fields. Then*

$$\sum_{\substack{\chi \in \widehat{G} \\ \chi \neq \mathbf{1}_G}} n(G, \chi)^2 \leq \left(\text{ord}_{s=s_0} \left(\frac{\zeta_E(s)}{\zeta_F(s)} \right) \right)^2.$$

Using ideas in the proof of Theorem 3.4 one can find and prove the right group theoretic lemma to prove the following theorem due to Murty and Raghuram. We refer the reader to [27] for more details on this theme.

Lemma 3.1. *Let G be a finite solvable group. Let H be a subgroup of G . Let $\{G^i\}_{i \geq 0}$ denote the derived series of G . Let k be the least non-negative integer such that $G^{k+1} = \{1\}$. (Assume that G has more than one element). Then, for all $i \geq 1$*

$$\text{Ind}_H^G(\mathbf{1}_H) = \text{Ind}_{H \cdot G^i}^G(\mathbf{1}_{H \cdot G^i}) + \sum_j \text{Ind}_{H_j}^G(\psi_j)$$

where ψ_j 's are one dimensional characters of some subgroups H_j of G and j runs over some indexing set which might possibly be empty and all of these depending on H and i .

Theorem 3.5. *Let G be the Galois group of a solvable Galois extension E/F of number fields. Let $[G, G]$ denote the commutator subgroup of G . Let E^{ab} denote the subfield of E/F fixed by $[G, G]$. Let S denote the set of distinct one dimensional characters of G . Then*

$$\sum_{\chi \notin S} n(G, \chi)^2 \leq \left(\text{ord}_{s=s_0} \left(\frac{\zeta_E(s)}{\zeta_{E^{\text{ab}}}(s)} \right) \right)^2.$$

We can use the above theorem and the Brauer - Aramata theorem and deduce the following corollary.

Corollary 3.7. *Let G be the Galois group of a solvable Galois extension E/F of number fields. Let E^{ab} be the fixed field of the commutator subgroup $[G, G]$. Let S denote the distinct one-dimensional characters of G . Then*

$$\prod_{\chi \notin S} L(s, \chi, E/F)^{\chi(1)}$$

is entire as a function of s and has only multiple zeros.

Proof : By the Artin-Takagi factorization (refer Theorem 2.3) the above product is exactly $\zeta_E(s)/\zeta_{E^{\text{ab}}}(s)$. By the Brauer-Aramata theorem (refer Corollary 3.1) this quotient is entire. Suppose this quotient has a simple zero then the right hand side of the inequality in Theorem 3.5 is 1. So there is at most one χ_0 such that $L(s, \chi_0, E/F)$ has a simple zero at $s = s_0$ and all the other $L(s, \chi, E/F)$ are necessarily non-vanishing at this point. But then since χ_0 is not in S we have $\chi_0(1) > 1$ and therefore the product above has a zero of order $\chi_0(1) > 1$ which contradicts the assumption that the quotient had a simple zero. Note that the corollary follows modulo the Artin conjecture. $\square$

4. SELBERG'S PROGRAM

4.1. The Selberg class. The Selberg class is a certain class of functions of one complex variable. The Riemann zeta function of Section 1.1 is a prototype of an element in the Selberg class. In this section we define the Selberg class and point out various examples of functions in the Selberg class.

Definition 4.1. *The Selberg class, denoted $\mathcal{S}$, consists of functions $F(s)$ of one complex variable $s = \sigma + it$ satisfying the following properties :*

1. **(Dirichlet Series)** *For $\sigma > 1$ the function $F(s)$ is represented by a Dirichlet series as $F(s) = \sum_{n \geq 1} a_n/n^s$. It is normalized by $a_1 = 1$. We also sometimes denote a_n by $a_n(F)$.*
2. **(Analytic continuation)** *The function $F(s)$ extends to a meromorphic function to the entire s -plane such that there exists an $m \geq 0$ for which $(s-1)^m F(s)$ is holomorphic of finite order.*
3. **(Functional Equation)** *There exists real numbers $Q > 0$, $\alpha_i > 0$ and complex numbers r_i with $\text{Re}(r_i) > 0$ such that, if*

$$\phi(s) = Q^s \left(\prod_{i=1}^d \Gamma(\alpha_i s + r_i) \right) F(s)$$

then the function ϕ satisfies the functional equation

$$\phi(s) = w \overline{\phi(1 - \bar{s})}$$

for some constant w of absolute value one.

4. **(Euler Product)** *The function $F(s)$ can be factored as*

$$F(s) = \prod_p F_p(s)$$

where p runs over all primes and the local factors F_p are of the form

$$F_p(s) = \exp\left(\sum_{k=1}^{\infty} b_{p^k}/p^{ks}\right).$$

Further, for some $\theta < 1/2$ one has $b_{p^k} = O(p^{k\theta})$.

5. **(Ramanujan Hypothesis)** *For any fixed $\epsilon > 0$, one has $a_n = O(n^\epsilon)$.*

We now take up the various L -functions in Section 1 and show that they belong to the Selberg class. We remark here that in all the known examples of functions in the Selberg class the local Euler factors $F_p(s)$ are known to be inverses of polynomials in p^{-s} with constant term 1. Some formal jugglery is enough to convince us that the Euler factor is

then of the form required in property (4) in the above definition. All the L -functions considered in Section 1 are defined as Dirichlet series and it is easily seen that they verify the Ramanujan Hypothesis. It is only the analytic continuation and functional equation one has to verify. These too follow from appropriate theorems in the corresponding sections.

Selberg introduced this class of functions to study the value distributions of linear combinations of Dirichlet series with Euler products and functional equations. See [32] for an elaboration on this theme.

4.2. Basic properties and the Selberg conjectures. In this section we give the basic properties of functions in the Selberg class and will introduce the Selberg conjectures, which predict a certain kind of orthonormal system in $\mathcal{S}$. These conjectures have rather remarkable consequences and this will be the theme of this and the following section.

In his paper [32] Selberg proves the following theorem on the number of zeros of a function in the Selberg class.

Theorem 4.1. *Let $F(s)$ be a function in the Selberg class. Let $N_F(T)$ be the number of zeros $\rho = \beta + i\gamma$ of F such that $0 < \gamma \leq T$. Then*

$$N_F(T) = \alpha_F \pi^{-1} T (\log T + c) + S_F(T) + O(1), \quad \text{as } T \rightarrow \infty$$

where c is a constant, $S_F(T) = O(\log T)$ and $\alpha_F = \sum_i \alpha_i$.

From the above theorem we get that the number $\alpha_F = \sum_i \alpha_i$ depends only on F . The number $d_F = 2\alpha_F$ is called the *dimension* of F . Selberg conjectures in [32] that the dimension d_F of any function $F \in \mathcal{S}$ is an integer. This question has been previously raised by Vignéras [39]. Note that since $N_{F_1 \cdot F_2}(T) = N_{F_1}(T) + N_{F_2}(T)$ we have that the dimension of $F_1 \cdot F_2$ is the sum of dimensions of F_1 and F_2 .

We can see, using the theorems in Section 1, that both the Riemann zeta function and the classical Dirichlet L -function are one dimensional. The Hecke L -function (and so also the Dedekind zeta function) associated to a Dirichlet character modulo an ideal $\mathfrak{m}$ of a number field K has dimension equal to the degree $[K : \mathbb{Q}]$ of K over $\mathbb{Q}$. From Section 2.1 it follows that for an irreducible character χ of $\text{Gal}(E/F)$ the Artin L -function $L(s, \chi, E/F)$ has dimension equal to $\chi(1)[K : \mathbb{Q}]$.

We now state an old theorem of Bochner [3] regarding the dimension of any function F in the Selberg class.

Theorem 4.2. *If $F \in \mathcal{S}$ and $\alpha_F > 0$ then $\alpha_F \geq 1/2$.*

We remark here that Conrey and Ghosh [9] give a simple proof of Bochner's theorem and they also show that the fact $b_{p^k} = O(p^{k\theta})$ with $\theta < 1/2$ (this being the essential point) implies that the only functions

in the Selberg class with dimension zero is the constant function 1. (Since we deal only with normalized Dirichlet series, i.e. $a_1 = 1$, the only constants in $\mathcal{S}$ is the constant function 1.)

We now go into the multiplicative properties of the Selberg class. We begin with a simple lemma.

Lemma 4.1. *The family $\mathcal{S}$ is multiplicatively closed.*

Definition 4.2. *A function $F \in \mathcal{S}$ is called **primitive** if the equation $F = F_1 \cdot F_2$ with F_1 and F_2 in $\mathcal{S}$ implies that either $F_1 = 1$ or $F_2 = 1$.*

We will discuss this notion more thoroughly below. Some natural questions to ask are whether there are any primitive functions at all and if there are then can every function in $\mathcal{S}$ be factored as a product of primitive functions. Can we make any uniqueness statements about such a factorization?

By Theorem 4.2 we have that any function in $\mathcal{S}$ which has dimension one must necessarily be primitive. So the Riemann zeta function and the classical Dirichlet L -function are primitive functions. We shall see later, assuming the Selberg conjectures, that the Artin L -functions corresponding to irreducible characters of Galois extensions of $\mathbb{Q}$ will be primitive. (So having dimension one is sufficient and is not in general necessary for a function to be primitive. The L -function attached to a Hecke eigenform (not of CM type) gives such an example. See [23].)

It is a theorem due to Selberg [32], Conrey and Ghosh [9] that it is possible to factorize any function in $\mathcal{S}$ into primitive functions. The dimension of F gives a convenient numerical invariant to prove such an assertion by an induction argument.

Theorem 4.3. *Every function in the Selberg class can be factorized into primitive functions.*

Proof : Take F in $\mathcal{S}$. If F is primitive then there is nothing to prove. If F is not primitive then write it as $F = F_1 \cdot F_2$ with both the F_i 's not being a constant. Then $d_{F_i} \leq d_F - 1$ and an induction argument finishes the proof. (Recall that $d_F = 1$ implies that F is primitive.) $\square$

We now go into the uniqueness of the factorization in Theorem 4.3. To this end we now introduce the Selberg conjectures.

Conjecture 4.1. (Selberg conjectures)

1. *For any primitive F in the Selberg class,*

$$\sum_{p \leq x} \frac{|a_p(F)|^2}{p} = \log \log x + O(1).$$

2. For two distinct primitive functions F and F' in the Selberg class

$$\sum_{p \leq x} \frac{a_p(F) \overline{a_p(F')}}{p} = O(1).$$

We now state and prove an easy theorem in [9] which says that the factorization of any $F \in \mathcal{S}$ into primitive functions is unique. To date there is no proof of unique factorization without assuming the Selberg conjectures.

Theorem 4.4. *Assume the Selberg conjectures, then :*

1. Any $F \in \mathcal{S}$ can be factored uniquely, upto a permutation of the factors, as a product of primitive functions.
2. Let $F \in \mathcal{S}$. Let $F = F_1^{e_1} \cdots F_k^{e_k}$ be the unique factorization of F . Let $n_F = \sum_i e_i^2$. Then

$$\sum_{p \leq x} \frac{|a_p(F)|^2}{p} = n_F \log \log x + O(1).$$

Proof : The proof of (1) follows from two basic observations. If F and G are in $\mathcal{S}$ then $a_p(F \cdot G) = a_p(F) + a_p(G)$ for any prime p . The second observation is that, assuming the Selberg conjectures, if P is a primitive function in $\mathcal{S}$ then the highest power (possibly zero) of P which occurs in the factorization of any $F \in \mathcal{S}$ is given by

$$\lim_{x \rightarrow \infty} \frac{\sum_{p \leq x} \frac{a_p(F) \overline{a_p(P)}}{p}}{\log \log x}$$

The rest of the details are left to the reader. The proof of (2) follows by taking ‘inner product’ of F with itself. $\square$

4.3. Selberg conjectures imply the Artin conjecture. The main aim of this section is to give a summary of the paper [22] due to the author in which are proved the two theorems presented here. The first says that the Selberg conjectures imply the Artin conjecture. Further if we consider solvable extensions of number fields then the Selberg conjectures imply the strong Artin conjecture.

Theorem 4.5. *The Selberg conjectures 4.1 imply the Artin conjecture 2.2.*

Proof : By Proposition 2.1 it is enough to consider the case when χ is an irreducible non-trivial character of G the Galois group of a Galois extension $K/\mathbb{Q}$. Using Brauer’s Theorem 3.1 and Artin’s reciprocity Theorem 2.6 we may write $L(s, \chi, K/\mathbb{Q})$ as L_1/L_2 where each L_i is a product of Hecke L -functions. Since Hecke L -functions are in the

Selberg class, assuming the Selberg conjectures, we have, using the unique factorization Theorem 4.4, that

$$L(s, \chi, K/\mathbb{Q}) = \prod_{i=1}^m F_i(s)^{e_i}.$$

Let $\chi(p)$ denote the p -th Dirichlet coefficient of $L(s, \chi, K/\mathbb{Q})$. Comparing the p -th Dirichlet coefficients on both sides of the above equation we get

$$\chi(p) = \sum_{i=1}^m e_i a_p(F_i).$$

From this we get, using the Selberg conjectures, that

$$\sum_{p \leq x} \frac{|\chi(p)|^2}{p} = \left(\sum_i e_i^2 \right) \log \log x + O(1)$$

We now make the following claim :

Claim :

$$\sum_{p \leq x} \frac{|\chi(p)|^2}{p} = \log \log x + O(1).$$

Assuming the claim we get that $\sum_{i=1}^m e_i^2 = 1$ which implies that $m = 1$ and $e_1 = \pm 1$. So $L(s, \chi, K/\mathbb{Q})$ is either $F(s)$ or $F(s)^{-1}$ for a primitive function $F(s)$ in the Selberg class. But since $L(s, \chi, K/\mathbb{Q})$ has trivial zeros, by property (2) of Definition 4.1, it must necessarily be $F(s)$. It now suffices to prove the claim.

The claim really follows from the irreducibility of χ and the Chebotarev density theorem. If C denotes a conjugacy class in G and g_C denotes any element in the conjugacy class C then we have

$$\sum_{p \leq x} \frac{|\chi(p)|^2}{p} = \sum_C |\chi(g_C)|^2 \sum_{p \leq x, \sigma_p \in C} 1/p.$$

The Chebotarev density theorem may be stated as

$$\sum_{p \leq x, \sigma_p \in C} 1/p = |C|/|G| \log \log x + O(1).$$

We therefore get using the irreducibility of χ

$$\begin{aligned} \sum_{p \leq x} \frac{|\chi(p)|^2}{p} &= \sum_C |\chi(g_C)|^2 |C|/|G| \log \log x + O(1) \\ &= (|G|^{-1} \sum_{g \in G} |\chi(g)|^2) \log \log x + O(1) \\ &= \log \log x + O(1) \end{aligned}$$

This proves the claim and as explained the theorem follows. $\square$

Corollary 4.1. *Assume the Selberg conjectures. Then*

1. *If χ is an irreducible non-trivial character of $\text{Gal}(K/\mathbb{Q})$ then the Artin L -function $L(s, \chi, K/\mathbb{Q})$ is a primitive element in the Selberg class.*
2. *Let χ be an irreducible non-trivial character of $\text{Gal}(E/F)$. In general $L(s, \chi, E/F)$ need not be a primitive element in the Selberg class. If the character obtained by inducing χ from $\text{Gal}(K/F)$ to $\text{Gal}(K/\mathbb{Q})$ is irreducible, where K is the Galois closure of $E/\mathbb{Q}$, then $L(s, \chi, E/F)$ is primitive as an element in the Selberg class.*

We now present the next main theorem of this section which states that the Selberg conjectures implies the strong Artin conjecture for solvable extensions of $\mathbb{Q}$.

Theorem 4.6. *Assume the Selberg conjectures. Then :*

1. *If π is an irreducible cuspidal automorphic representation of $GL_n(\mathbb{A}_{\mathbb{Q}})$ that satisfies the Ramanujan Hypothesis then $L(s, \pi)$ is in the Selberg class and is primitive as an element of this class.*
2. *Let $K/\mathbb{Q}$ be a solvable extension. Let χ be an irreducible character of $\text{Gal}(K/\mathbb{Q})$ of degree n . Then there is a cuspidal automorphic representation π of $GL_n(\mathbb{A}_{\mathbb{Q}})$ such that $L(s, \pi) = L(s, \chi, K/\mathbb{Q})$.*

Proof : Proof of (1) is non-trivial and uses the works of Jacquet-Shalika, Shahidi and Arthur-Clozel. See [22] for details of this. (Statement (1) is recorded as Lemma 4.2 in [22].) Assuming (1), a sketch of the proof of (2) is as follows.

Since $K/\mathbb{Q}$ is solvable, by Arthur-Clozel [1], the Dedekind zeta function ζ_K is an automorphic L -function over $\mathbb{Q}$. Then by [1] there are finitely many irreducible cuspidal automorphic representations π_i 's of $GL_{m_i}(\mathbb{A}_{\mathbb{Q}})$ such that

$$\zeta_K(s) = \prod_i L(s, \pi_i).$$

Now using the Artin-Takagi factorization (cf. Theorem 2.3) we get

$$\prod_i L(s, \pi_i) = \prod_{\chi \in \widehat{G}} L(s, \chi, K/\mathbb{Q})^{\chi(1)}$$

where $G = \text{Gal}(K/\mathbb{Q})$. The above equation forces the Ramanujan Hypothesis on all the $L(s, \pi_i)$. Since the right hand side of the above equation is $\zeta_K(s)$ whose eigenvalues of the p -Euler factors are of absolute value one, so are the eigenvalues of the p -Euler factors of the left hand side, which is the content of the Ramanujan Hypothesis. We now get a proof of Statement (2) by using Statement (1), Corollary 4.1 and Theorem 4.4. $\square$

Remark 4.1. Note that in the proofs of Theorems 4.5 and 4.6 we have only used the unique factorization in the Selberg class and haven't really used the statement of the Selberg conjectures themselves. (Of course, Theorem 4.4 allows us to do so.) This remark will be useful in Section 5.2.

5. PAIR CORRELATION

5.1. The classical pair correlation. In this section a short summary of Montgomery's paper [19] is presented. In this paper he studies the distribution of pairs of zeros of the Riemann zeta function of Section 1.1.

Throughout this section we assume the Riemann Hypothesis, henceforth abbreviated as (RH), which is stated as Conjecture 1.1. So let $\rho = 1/2 + i\gamma$ be a non-trivial zero of $\zeta(s)$. The aim is to study the distribution of differences $\gamma - \gamma'$ for zeros $1/2 + i\gamma$ and $1/2 + i\gamma'$. To this end Montgomery defines the following function.

$$(9) \quad F(\alpha) = F(\alpha, T) = \left(\frac{T \log T}{2\pi} \right)^{-1} \sum_{0 < \gamma, \gamma' \leq T} T^{i\alpha(\gamma - \gamma')} w(\gamma - \gamma')$$

where α and T are real and $T \geq 2$. The function w is a suitable weight function which is taken as

$$w(u) = \frac{4}{4 + u^2}.$$

To investigate sums involving $\gamma - \gamma'$ one has to convolve $F(\alpha)$ with a suitable function $\widehat{g}(\alpha)$. From Equation 9 using elementary properties of Fourier transforms, we get the following equation.

(10)

$$\sum_{0 < \gamma, \gamma' \leq T} r \left((\gamma - \gamma') \frac{\log T}{2\pi} \right) w(\gamma - \gamma') = \frac{T \log T}{2\pi} \int_{-\infty}^{\infty} F(\alpha) \hat{r}(\alpha) d\alpha.$$

In the above equation $\hat{r}(\alpha)$ is the Fourier transform of r and is given by :

$$\hat{r}(\alpha) = \int_{-\infty}^{\infty} r(u) e^{-2\pi i u \alpha} du.$$

Basically given a set S in $\mathbb{R}$ and given a weight function w on S (which extends to a nice function on $\mathbb{R}$) we get a distribution on $\mathbb{R}$ by $r \mapsto \sum_{s \in S} r(s) w(s)$. The Fourier transform of this distribution is given by integrating against a function, say F . So this F is characterised by the property that

$$\sum_{s \in S} r(s) w(s) = \int_{\mathbb{R}} F(x) \hat{r}(x) dx$$

for all r in a suitable (Schwartz) space of functions on $\mathbb{R}$. Indeed, looking at

$$F(\alpha) = \sum_{s \in S} w(s) e^{i\alpha s},$$

we have that

$$\int_{\mathbb{R}} F(\alpha) \hat{r}(\alpha) d\alpha = \int_{\mathbb{R}} \left(\sum_{s \in S} w(s) e^{i\alpha s} \right) \hat{r}(\alpha) d\alpha$$

from which the penultimate identity is clear. In the case at hand the set S is the set of differences of imaginary values of non-trivial zeros of the zeta function. To make integrals converge, one looks at only those zeros whose imaginary values are non-negative and bounded above by T . To make allowances for the vagaries of the zeta function an appropriate $T \log T$ is thrown in. The point now is to exploit the two Equations 9 and 10. One of the main theorems of [19] is the following.

Theorem 5.1. *Assume (RH). Let α and T be real. Let $T \geq 2$. Let $F(\alpha) = F(\alpha, T)$ be as in Equation 9. Then $F(\alpha)$ is real and $F(\alpha) = F(-\alpha)$. For fixed α such that $0 \leq \alpha < 1$ we have*

$$F(\alpha) = (1 + o(1)) T^{-2\alpha} \log T + \alpha + o(1)$$

as $T \rightarrow \infty$. This is uniform for $0 \leq \alpha \leq 1 - \epsilon$.

Using the above theorem and suitable test functions r we get the following corollary.

Corollary 5.1. *Assume (RH). If $0 < \alpha < 1$ is fixed then*

1.

$$\sum_{0 \leq \gamma, \gamma' \leq T} \frac{\sin(\alpha(\gamma - \gamma') \log T)}{\alpha(\gamma - \gamma') \log T} w(\gamma - \gamma') \sim \left(\frac{1}{2\alpha} + \frac{\alpha}{2} \right) \frac{T \log T}{2\pi}.$$

2.

$$\sum_{0 \leq \gamma, \gamma' \leq T} \left(\frac{\sin(\alpha(\gamma - \gamma') \log T)}{\alpha(\gamma - \gamma') \log T} \right)^2 w(\gamma - \gamma') \sim \left(\frac{1}{\alpha} + \frac{\alpha}{3} \right) \frac{T \log T}{2\pi}.$$

Statement (2) of the previous corollary is used to prove the fact that at least two thirds of the zeros of the Riemann zeta function are simple. (This is remarkable because before Montgomery's paper the only result in this direction was due to Selberg [31] which appeared almost thirty years before [19]. Selberg proved (unconditionally) that a positive density of zeros of the Riemann zeta function have an odd order and lie on the critical line.) This fact is recorded in the corollary given below. (Recall that the number of zeros of the Riemann zeta function whose imaginary value lies between 0 and T behaves like $T \log T / 2\pi$.) It is expected that all the zeros of the zeta function are simple.

Corollary 5.2. *Assume (RH).*

$$\sum_{\substack{\rho \text{ simple, } 0 \leq \gamma \leq T}} 1 \geq \left(\frac{2}{3} + o(1) \right) \frac{T \log T}{2\pi}.$$

Note that in Theorem 5.1 the formula describes the behaviour of F when $\alpha < 1$. One may ask for the behaviour of F when $\alpha \geq 1$. Some easy remarks (see page 183 of [19]) suggests that Theorem 5.1 will no longer be valid in this region. Some heuristics (see Section 3 of [19]) suggest the following conjecture, which we call *the classical pair correlation conjecture*.

Conjecture 5.1. *Assume (RH). Let $\alpha \geq 1$ be fixed. Then*

$$F(\alpha) = 1 + o(1)$$

as $T \rightarrow \infty$.

It can be seen, using Equation 10, that the above conjecture is equivalent to the fact that Equation 11 below holds. For any real α and β let $I(\alpha, \beta)$ be the set $X \in \mathbb{R}$ such that $2\pi\alpha/\log T \leq x \leq 2\pi\beta/\log T$.

(11)

$$\sum_{\substack{0 \leq \gamma, \gamma' \leq T \\ \gamma - \gamma' \in I(\alpha, \beta)}} 1 \sim \left(\int_{\alpha}^{\beta} \left(1 - \left(\frac{\sin \pi u}{\pi u} \right)^2 \right) du + \delta(\alpha, \beta) \right) \frac{T \log T}{2\pi}.$$

Here $\delta(\alpha, \beta)$ is 1 if α and β have opposite signs and is 0 otherwise.

In a certain terminology the above equation is formulated as the assertion that the function $1 - (\sin \pi u / \pi u)^2$ is the *pair correlation function* of the zeros of the zeta function. It follows easily from the above equation that all zeros of the zeta function are simple. (Take β positive and close to zero and let $\alpha = -\beta$.)

In [19] is mentioned a remark due to F. J. Dyson that the eigenvalues of a random complex Hermitian or a unitary matrix of large order has exactly the same pair correlation function. This suggests that there maybe a linear operator whose eigenvalues are the zeros of the zeta function.

5.2. Pair correlation for the Selberg class. In this section we present a summary of the paper [26] due to author and Perelli, which formulates an appropriate version of pair correlation of functions in the Selberg class $\mathcal{S}$ which was considered in Section 4.

Here we take two functions F and G in the Selberg class $\mathcal{S}$ and compare the zeros of F against the zeros of G . (We continue to use the notation set up in Section 4 regarding all matters concerning the Selberg class, in particular for the data defining a function $F \in \mathcal{S}$.)

We also assume the *Generalized Riemann Hypothesis*, henceforth abbreviated (GRH) for the Selberg class. Recall that the trivial zeros of a function $F \in \mathcal{S}$ are those which come from poles on the Γ -factors in its functional equation. By (GRH) we mean that all the non-trivial zeros of any $F \in \mathcal{S}$ occur on the line $\sigma = 1/2$.

Let $F, G \in \mathcal{S}$. To this is attached the following function.

$$(12) \quad \mathcal{F}_{F,G}(\alpha) = \frac{\pi}{d_F T \log T} \sum_{-T \leq \gamma_F, \gamma_G \leq T} T^{i\alpha d_F(\gamma_F - \gamma_G)} w(\gamma_F - \gamma_G)$$

where d_F is the dimension of F , $1/2 + i\gamma_F$ is any (non-trivial) zero of F and similarly for G and further w is the same weight function as in Section 5.2. The similarity with Equation 9 is obvious.

Note that the function $\mathcal{F}_{F,G}$ is not symmetrical in F and G . This although not absolutely necessary turns out to be technically useful. (Refer [26] where they fix F and ‘play’ G against F .)

The pair correlation conjecture for the Selberg class can now be stated.

Conjecture 5.2. *Let F and G be two primitive functions in $\mathcal{S}$. Then*

$$\mathcal{F}_{F,G}(\alpha) = \begin{cases} \delta_{F,G}|\alpha| + d_G T^{-2|\alpha|d_F} \log T(1 + o(1)) + o(1) & , \text{ if } |\alpha| < 1 \\ \delta_{F,G}|\alpha| + o(1) & , \text{ if } |\alpha| \geq 1 \end{cases}$$

as $T \rightarrow \infty$ and uniformly for α in any bounded interval. The symbol $\delta_{F,G}$ stands for 1 if $F = G$ and for 0 otherwise.

The non-trivial zeros of F and G are said to be *strongly distinct* if they are placed on different points on the line $\sigma = 1/2$. One of the main theorems proved in [26] is the following one where some consequences of the pair correlation conjecture for the Selberg class are presented. In particular one has that the pair correlation conjecture implies the Selberg conjecture 4.1. Using Theorem 4.5 we therefore get that the pair correlation conjecture implies the Artin conjecture 2.2.

Theorem 5.2. *Let F and G be two distinct functions in the Selberg class $\mathcal{S}$.*

1. *Assume Conjecture 5.2. Then all but $o(T \log T)$ non-trivial zeros upto T of $F(s)$ are simple.*
2. *Assume Conjecture 5.2. All but $o(T \log T)$ non-trivial zeros upto T of $F(s)$ and $G(s)$ are strongly distinct.*
3. *If the asymptotic formula in Conjecture 5.2 holds for some α_0 then $\mathcal{S}$ has unique factorization.*

We remark that Statement (2) of the above theorem should be compared with Theorem 1 of Bombieri-Perelli [4] where the pair correlation conjecture is not assumed.

Corollary 5.3. *The pair correlation conjecture 5.2 for the Selberg class implies the Artin conjecture 2.2.*

Proof : Follows from Statement (3) of Theorem 5.2, Remark 4.1 and Theorem 4.5. $\square$

We remark here that by Corollary 3.6, using the generalized Riemann hypothesis for ζ_E , we get for an irreducible non-trivial character χ of $\text{Gal}(E/F)$ that the Artin L -function $L(s, \chi, E/F)$ is entire in the half plane $\sigma > 1/2$. Pair correlation is needed to deal with the *critical line* $\sigma = 1/2$. Of course, if $L(s, \chi, E/F)$ is holomorphic on $\sigma \geq 1/2$ then using the functional equation in Theorem 2.1 (and the fact that the Γ function is nowhere vanishing) we get the Artin conjecture.

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